

MIE245 - Winter 2026

Lab 3 - Quick Sort & Radix Sort

Deadline: Tuesday, January 27th, 11:59pm EST

Submission Instructions

Reminder: You can not claim work that was completed by others or generative AI as your own. Submit a completed **readme** template file outlining what code you contributed and what code was completed by other students, people, tools, and/or leveraged from websites.

Add `_utorid` (with `utorid` replaced by your UTORid in all lower-case characters) to the end of both the completed template file names (i.e., the `.py` and the `readme` files). Please ensure you use your UTORid (i.e., what you use to sign into Quercus and **not** other identifiers such as your student number).

Compress your completed template files to a zip file and upload the zip file to Quercus.

Lab Instructions

In this lab, you will implement the Quick Sort and Radix Sort algorithms from Lecture 8 (see slides).

Complete the template file **MIE245_Lab_3_Template.py** by finishing the following required methods:

- `radix_sort(array, base=10)`: performs Radix Sort on the input array, returns the sorted array
- `quick_sort(array)`: initializes Quick Sort, returns the sorted array
- `partition_and_swap_pivot(array, lo, hi)`: identifies the pivot element and swaps it to `lo`, returns the pivot element. For pivot selection, use the Naive approach (Lecture 8, slide 7).
- `sort_aux(array, lo, hi)`: performs Quick Sort on the input array, returns the sorted array

AutoGrading:

You can access an example of the auto-grading script that will be used to test your submission (**MIE245_Lab_3_Autograder.py**). Please follow the steps below to run the auto-grader for the provided test case:

1. Place the auto-grader script in a new folder
2. Inside the newly created folder, create a folder called “submissions”
3. Place your script in the “submissions” folder

4. Run the auto-grader script and open the **grades.csv** file to see if you passed the test case (1 is pass, 0 is fail)

Note: The auto-grading script only contains one of the several test cases we will evaluate against, so even if you pass this test case, you may not have a completely correct solution.

Tip: You can modify the test case to test additional functionality.